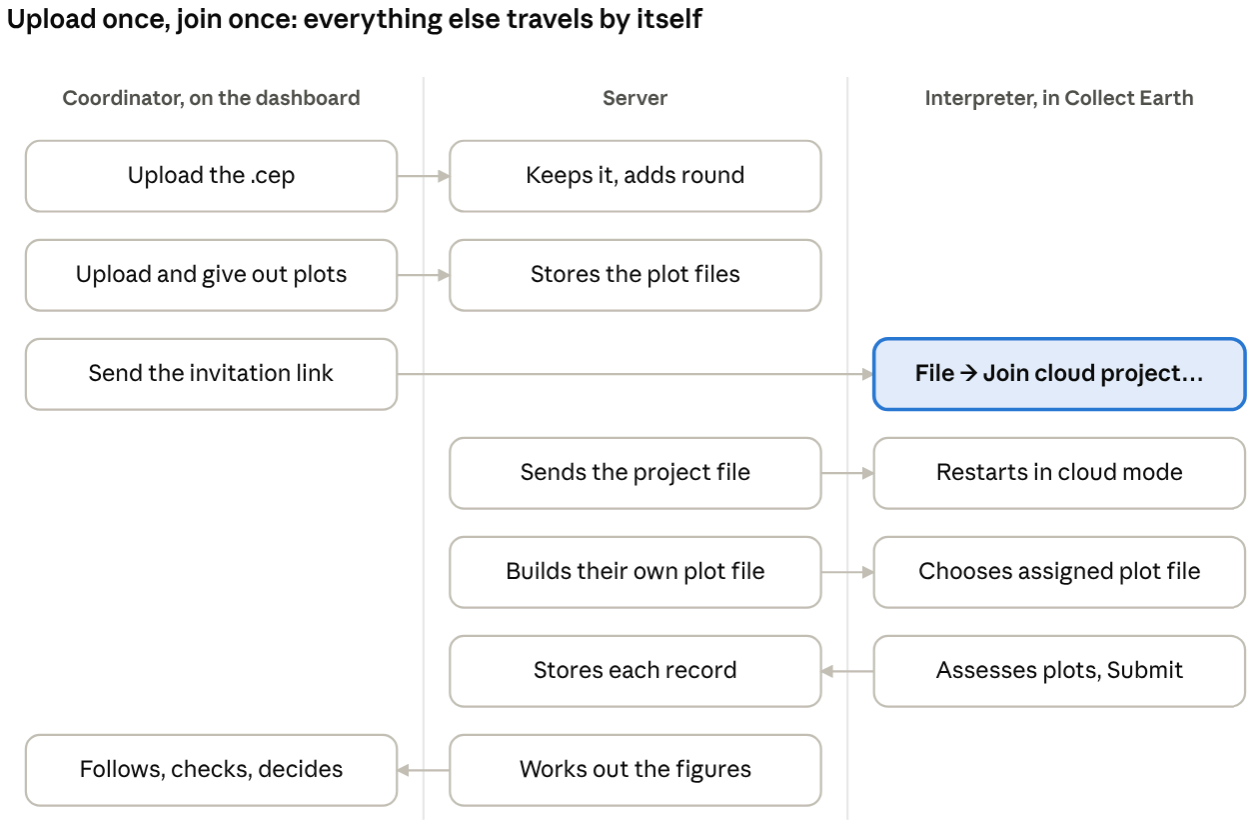
How the survey reaches the interpreters in a cloud project

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In a cloud project the coordinator puts the survey and the plots on the server once. Each interpreter joins once with an invitation link, and Collect Earth then fetches the project file and their own plots by itself, and sends every saved plot back within 30 seconds.

At a glance



the journey of the survey · 7 steps, 3 places

Read it top to bottom: steps 1 to 3 are the coordinator's, step 3 is the interpreter's only manual step besides choosing a plot file, and steps 6 and 7 repeat for every plot.

What is shared

In a cloud project the coordinator never sends files by email. They upload two kinds of file to the server once; each interpreter's Collect Earth downloads them by itself. The plots then travel back the same way.

File	Made with	Uploaded on the dashboard	Who gets it, and when
Project file (`.cep`)	Collect Earth's survey designer (or Collect)	**Project file** tab	Every member, when they join the project
Plot files (`.csv`)	A spreadsheet or Collect Earth's sampling tools	**Plot files** tab	Only the people given each file, or their share of it, when they choose it in Collect Earth
Records (one per plot and person)	Collect Earth, as people work	Not uploaded by hand	Sent to the server every 30 seconds

What is inside the project file

A `.cep` file is a ZIP. It holds:

- the **survey** (`placemark.idm.xml`): the questions, their answers and the plot key, usually the plot `id`;
- the **form** of the plot (`balloon.html`) and the Google Earth map template (`kml_template.fmt`);
- the **project settings** (`project_definition.properties`): the survey name, plot shape, sampling points, imagery and language;
- usually a sample plot file, a survey guide (PDF) and supporting files.

The plot file inside the `.cep` is only what Collect Earth shows right after joining: each person then works on the plot files the coordinator gives them.

What the server keeps, and what it changes

The server keeps **the file you uploaded** and, when needed, **a version it hands out**. It changes the survey in two cases, only while no plot has been collected:

1. **A hidden `round` key.** When the survey's only key is the plot `id`, the server adds a second key called `round`. Round 1 is a plot's first assessment, round 2 a quality check by another person. Collect Earth then saves a check as a separate record (`17,1` and `17,2`), so a check never overwrites the first assessment. Interpreters never see the round.
2. **A coordinator's comment field** (`coordinator_note`). When a coordinator sends a plot back for another look, the comment appears at the top of that plot's form, read-only.

The **Project file** tab says what was done, and why when it could not be (for example a survey with several keys). The file you uploaded can always be downloaded again from there. A survey that already has a `round` key keeps it: you then choose which of its codes marks a re-assessment, or *None* when `round` means something else, such as a year.

What never travels

- Nobody receives anyone else's plots or answers. Each person's Collect Earth gets only their own plot file, or their share of a divided file.
- Interpreters cannot tell which of their plots are quality checks or reference plots: those arrive in the same plot file as any other plot.
- Training plots go to a separate part of the server and never mix with the project's data.

Step 1, the coordinator: putting the files on the server

The project file

1. Open the project on the dashboard, then the **Project file** tab.
2. Drag the `.cep` file onto the page, or press **Choose the .cep file**. A file that is not a `.cep` is refused at once.
3. The server stores it and checks the survey inside. The page then says:
   - *Project file uploaded ({size}), last updated {date}*;
   - under **Assessment round**, whether the hidden `round` was added, or why not;
   - whether coordinators' comments will show in Collect Earth's form.
4. From now on, everyone who **joins** gets this version.

What the server refuses, and why:

Message	When	What to do
This file is too large for the server	The file, unzipped, is too big	Remove large extras (videos, big images) from the `.cep`
There is not enough space left in this project	The project is over its storage limit	Ask the server's administrator for more space
This project file has different keys from the one the data was collected with	Plots were already collected, and the new survey's keys differ	Upload a version with the same keys

Message	When	What to do
This project file has no 'round', but the data already collected uses it	The new file dropped the `round` key	Upload a version that keeps `round`
…lacks one of them (round codes 1 and 2)	The new file dropped a `round` code in use	Keep both codes
This project file reads the plots in a different way…	The new file would change how divided plot files are read	Set each divided file back to *Whole file*, upload, then divide again

The first two can happen at any time. The last four only protect data already collected: before any plot arrives, any survey can be uploaded.

The plot files

1. Open the **Plot files** tab and drag in one or more `.csv` files, or press **Choose CSV files**.
2. Each file must have, on every row, the plot id, then latitude and longitude, then any other columns the survey reads from the plot file. These are the same files Collect Earth opens locally.
3. The server reads each file as Collect Earth will, and reports *{name} uploaded: {count} plots*, with warnings when:
   - a plot id is on more than one row (Collect Earth takes them as one plot);
   - a row has no usable latitude and longitude (Collect Earth cannot place it);
   - a plot id does not fit the survey (for example letters where the survey takes numbers: Collect Earth would not find the records of those plots).
4. **Same name = new version.** A file with the name of one already there replaces it. If the new version leaves plots out, the dashboard first shows how many are given to people and how many have submitted records, and asks before replacing.
5. A plot file is refused if it holds plots of a **training set**: those stay apart.

Giving the plots out

Nothing reaches an interpreter until a file is given to them. Either:

- **Whole file:** in the table **Who collects each file**, tick a box for each person and file, then **Save**. Each person gets the whole file.
- **Divided:** press **Divide plots** on a file, choose **Evenly between people** or **By a column** (a province, a stratum), the order (*In the order of the file*, *Mix them*, or *Keep plots that are close on the map together*), and optionally a share of plots (5 to 50 %) also given to

a second person, blind, to compare answers. **Preview**, then **Apply**. Each person then gets a file with only their plots.

The server builds each person's file when they ask for it: the same header, their rows only, and the hidden round column where the survey has one (1 for their own plots, 2 for their check copies).

Step 2, the interpreter: joining the project

Joining is the only time the project file travels to an interpreter by itself. It happens once per person.

The invitation link

The coordinator makes a link on the **People** tab (a role, how many people may use it, how long it works) and sends it. It looks like `https://<server>/join/<code>`. The link carries the server's address, so nobody types it.

Opened in a browser, the link shows a page with the project's name and role, and the steps below. It explains that the collecting is done in Collect Earth, not in the browser. An expired, stopped or used-up link says so, and the person asks for a new one.

In Collect Earth

1. **File → Join cloud project…** opens *Join cloud project*.
2. Paste the link under **Invitation link** and press **Check the invitation**. Collect Earth asks the server what the link is for.
3. It shows: *You are about to join the cloud project "{name}" as {role}. Collect Earth will download the project and switch to cloud mode.* Press **Continue**.
4. The **Cloud login** dialog asks for a **Username** and **Password**:
   - **Register** creates a new account with the invitation (the first time on this server).
   - **Log in** uses an account the person already has on this server.
5. Collect Earth then, in this order:
   a. signs in and keeps the session;
   b. downloads the project file as the server hands it out (with the hidden `round`, where added);
   c. opens it as a project, exactly like **File → Import CEP file**: it is unpacked into the user folder (`projects/<survey name>`);
   d. saves the server's address, the project's id and the session in its settings, and sets the account name as the operator name;
   e. switches **Data storage** to *Cloud project (local database + automatic sync)*.

6. Collect Earth **restarts**. A small window shows each step until the main window opens. The title bar ends with *[Cloud project]* and the name in the top bar is the account.

If any step fails (a wrong password, no connection), nothing is changed and the error is shown in the dialog.

What changes on the interpreter's computer

	Before joining	After joining
Project	Whatever was open	The cloud project's survey
Data storage	SQLite, PostgreSQL or another cloud project	Cloud project: the same local SQLite file, plus sync
Operator name	Typed by the person	The account name, which cannot be changed in Collect Earth
Plots	Any plot file	Only the files the coordinator gives them (next step)

Records already in the local database stay there. Only records of this project's survey are sent to the server.

Logging in again later

The session lasts until the password changes or the person logs out. When Collect Earth starts without a valid session, it asks: *You are not logged in to the cloud project. Log in now to identify yourself and sync your data?* Until they log in, plots are saved locally and wait in the queue; nothing is lost.

The survey definition

When a **coordinator** runs Collect Earth in the project, it also sends the survey's definition to the server once, so the server can read records and build downloads. Collectors' and reviewers' Collect Earth skips this: the server refuses it from them, by design.

Step 3, the interpreter: opening their plots, and sending them back

Choosing the assigned plot file

1. In Collect Earth, open **Settings**. In a cloud project, the **Project** section shows **Assigned plot file**: the files the coordinator gave this person, from the server.

2. Choose one. Collect Earth downloads it, checks it against the server's checksum, and keeps it in an `assignments` folder next to the project. It becomes the project's plot file.
3. **Apply**. Collect Earth loads the plots; no restart is needed.

When nothing is given to them yet, the list says *The cloud server assigns you no plot file in this project yet*.

What the file holds

The coordinator gave them	The file they get
A whole file	The whole file
A share of a divided file	The same header and only their rows, built by the server when they ask
Check copies (overlap, automatic checks, reference plots)	The same file: those rows look like any other plot, with the hidden round 2
A training set	A separate file named *Calibration – {name}* or *Learning – {name}*

The round column is filled in by the server: 1 for a person's own plots, 2 for their check copies. Collect Earth saves a check as its own record, so it never overwrites anyone's first assessment. The interpreter cannot tell which plots are checks.

When the coordinator changes what they have

A new division, moved plots, new checks or new reference plots change the person's file on the server, not on their computer. They press **Refresh assignments** in **Settings**, choose the file again, and **Apply**. Plots they already submitted keep their records either way.

Sending the plots back

Every save of a plot (each answer, and **Submit**) is written to the local database first, then queued for the server:

- **Every 30 seconds** Collect Earth sends what is queued, up to 50 records at a time, compressed.
- **Every 10 minutes** it compares the local database with what was sent, and queues anything missed.
- **When Collect Earth quits**, it tries once more for up to 10 seconds.

- **Offline**, the queue keeps everything and retries, waiting longer each time, up to once an hour. Working offline for days is fine.

The server keeps one record per plot, round and person. A newer save replaces an older one; an older save arriving late is ignored (*stale*); two people on the same plot never overwrite each other.

File → Cloud sync status… shows *Waiting to sync*, *Failed (will retry)*, *Synced*, the *Last successful sync* and the *Last error*, with a **Sync now** button.

When the survey or the plots change during the work

Plot files reach people again whenever they refresh their assignments. The project file does not: Collect Earth downloads it only when joining. This difference matters most once people are working.

What changes	What reaches the interpreters by itself	What they have to do
A plot file replaced (same name)	The new rows, the next time they ask	**Refresh assignments**, choose the file, **Apply**
A new division, moved plots, new checks or reference plots	Their new share, the next time they ask	The same
A new plot file given to them	It appears in their list	The same
The project file replaced	**Nothing**	Join again with a *link for the new project file* (below)
A new attempt of a learning set	A note on **My work**	Choose the training file again: Collect Earth clears their earlier answers on those plots so they start afresh

A new version of the project file

1. The coordinator uploads it on the **Project file** tab (**Choose a new version**). The server checks it still matches the plots already collected (the same keys, `round` and its codes kept, plots read the same way), and refuses it otherwise.
2. From then on, people who **join** get the new version.

3. People who joined before keep the old one. Their page on the **People** tab says *Has an old project file: it was replaced after their last download*. On **My work** they see *Your project file is old*.
4. For each of them, the coordinator presses **Make a link for the new project file** on their page and sends it to that person only.
5. In Collect Earth, they choose **File → Join cloud project…**, paste it, confirm, and press **Log in** with their own account, **not Register**. Collect Earth downloads the project file as it is now and restarts. Their role and their plots do not change.

Records already saved stay, in the local database and on the server. They keep the answers they were saved with: a question added later is empty on them, and a removed question's answers stay in the record.

Good practice

- Settle the survey before inviting people: test the `.cep` locally, then upload it once.
- If a change is unavoidable, add questions rather than change or remove them, and keep the keys exactly as they are.
- Ask everyone to sync (**Cloud sync status… → Sync now**) before they join again with the new link.
- Tell people when you change their plots, so they press **Refresh assignments**.

Problems, and what is kept where

Common problems

What the person sees	Why	What to do
The invitation page says *This invitation link does not work*	The link expired, was stopped, or was used by as many people as allowed	The coordinator makes a new one
Login failed when joining	Wrong password, or **Register** pressed for an existing username	Use **Log in** with the existing account, or another username
After joining, the map shows the sample plots of the project file	No assigned plot file chosen yet	**Settings → Assigned plot file**, then **Apply**

What the person sees	Why	What to do
The cloud server assigns you no plot file in this project yet	Nothing was given to them, or their share of a divided file is empty	The coordinator gives them plots; they press **Refresh assignments**
New plots the coordinator gave do not appear	Collect Earth keeps the file it downloaded	**Refresh assignments**, choose the file, **Apply**
The form lacks a question the coordinator added	The project file was replaced after they joined	Join again with a link for the new project file, pressing **Log in**
Cloud sync status shows plots waiting	No connection, or the server is unreachable	Nothing: they are sent once it is back. **Sync now** tries at once
You are not logged in to the cloud project	The password changed, or the session ended	Log in again; the queued plots are then sent
The coordinator sees no plots from someone	They have not synced, or are not logged in, or joined another project	Ask them to open **Cloud sync status…** and read it out

What is kept where

	On the interpreter's computer	On the server
Project file	The version downloaded when they joined, unpacked in the user folder	The file uploaded and the version handed out
Plot files	The files they chose, in the project's `assignments` folder	Every plot file, and who has which plots
Records	All of theirs, in the local SQLite file, and in its automatic backups	One per plot, round and person, with every change logged
Sync queue	What is still to send	—
Login	The session, in Collect Earth's settings	The account

Because every plot is saved locally first, a lost connection or a server problem never loses work: the queue sends it later. A computer lost before syncing does lose its unsynced plots, so people should not work offline for weeks.